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Computer tablet distraction reduces pain and anxiety in pediatric burn patients undergoing hydrotherapy: A randomized trial

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ABSTRACT

Background: Distraction is often used in conjunction with analgesics to minimize pain in pediatric burn patients during treatment procedures. Computer tablets provide many options for distraction items in one tool and are often used during medical procedures. Few studies have examined the effectiveness of tablet distraction in improving the care of pediatric burn patients.
Aim: This study examines the effectiveness of tablet distraction provided by a child life specialist to minimize pain and anxiety in pediatric burn patients undergoing hydrotherapy.
Methods: Thirty pediatric patients (4–12) undergoing hydrotherapy for the treatment of burns participated in this randomized clinical trial. The tablet distraction group received tablet distraction provided by a child life specialist while those in the control group received standard care. Pain was assessed through self-reports and observation reports. Anxiety was assessed through behavioral observations. Length of procedure was also recorded.
Results: Nurses reported significantly less pain for the tablet distraction group compared to the control group. There was no significant difference between groups on self-reported pain. The tablet distraction group displayed significantly less anxiety during the procedure compared to the control group. Also, the tablet distraction group returned to baseline after the procedure while those in the control group displayed higher anxiety post-procedure. There was no difference in the length of the procedure between groups.
Conclusions: These findings suggest tablet distraction provided by a child life specialist may be an effective method for improving pain and anxiety in children undergoing hydrotherapy treatment for burns.

Abbreviations: TBSA, total body surface area affected; MMD, multi-modal distraction; IV, intravenous; CEMS, children's emotional manifestation scale; CCLS, Certified Child Life Specialist.

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1. Introduction

Pediatric burn patients experience physical and psychosocial distress during the initial trauma and during the ongoing treatment of the burns [1]. Much of this ongoing distress is due to procedure pain, as it is noted as the most severe and common type of pain for burn patients [2]. Unfortunately, procedure pain is also most likely to be viewed as undertreated, with pharmacological methods often failing to meet children's needs for pain management [3,4]. However, when nonpharmacological methods are used in conjunction with pharmacological ones, pain during burn procedures is significantly reduced [5,6].

One type of nonpharmacological support offered to pediatric burn patients is distraction. Distraction is a type of procedure support in which the attention of children undergoing medical procedures is focused away from the procedure, pain, or thoughts of pain/distress to a more neutral stimulus [7]. Distraction activities such as bubbles, I-Spy, music, and video goggles have been found to decrease distress and pain in children undergoing medical procedures [1,8,9].

Virtual reality, multi-modal distraction (MMD), and interactive gaming consoles are a few distraction tools that utilize technology to improve pediatric experiences for children undergoing burn treatments [10–12]. Such an activity draws the children's attention into the cognitive and sensory stimulation of the animation or game and limits the cognitive resources available to think about or process pain of the procedure (i.e., distraction) [6,13]. In fact, fMRI scans show decreased activity in pain related brain activity during virtual reality [13]. For children undergoing treatments for burns, technology based distractions have been found to be an effective nonpharmacological adjunctive pain treatment during physical therapy [1,6], wound care [14], and rehabilitation exercise therapy [12].

One popular technology based distraction tool is a computer tablet (i.e. tablet). With the touch of a screen, pediatric patients have access to movies, music, interactive games, breathing tools, books, puzzles, and more. Using their assessment of a child's development and needs, as well as the upcoming procedure, medical team members, such as child life specialists, provide possible distraction items on the tablet to help their patient's cope during procedures. In other words, a tablet caters to a wide variety of developmental and stress potential needs of patients through one tool. This is important because distraction activities work best when the activity is age appropriate and intriguing to the child [15].

There is growing use of tablets in pediatric facilities because of the accessibility and flexibility of these devices. However, there is little research examining the effectiveness of tablets in improving pediatric patient care. A review of the literature finds a handful of studies that have examined the effectiveness of tablets for pediatric patients. Positive benefits of tablets during pediatric procedures have been found, including a decrease in perioperative anxiety [16], immunization pain [17], and anxiety during emergency room visits [18]. These findings suggest that tablets are effective in minimizing pain and distress in children undergoing procedures.

To date, studies examining the effectiveness of tablets in minimizing pain and distress in pediatric burn patients undergoing treatment procedures seem rare. The aim of this study was to examine the effectiveness of a computer tablet distraction provided by child life specialists for decreasing pain and anxiety in pediatric burn patients undergoing hydrotherapy. We hypothesized that children who experienced distraction by a child life specialist using a tablet would display less self-reported and observed pain as well as fewer behaviors indicative of anxiety while undergoing hydrotherapy compared to children who did not receive tablet distraction during hydrotherapy.

2. Methods

This study utilized a randomized control trial design. Ethics approval for this study was obtained from the Institutional Review Board at the University of Alabama at Birmingham and the University of Alabama.

2.1. *Participants*

Participants were recruited from the Children's of Alabama Burn Center in Birmingham, Alabama by one of the primary researchers who is a child life specialist. All subjects were pediatric inpatients admitted for the treatment of a new burn medically utilising hydrotherapy as standard treatment. Eligible participants were between the ages of 4 and 12 and were undergoing a hydrotherapy procedure with standard intravenous analgesia for pain management. For consistency purposes, only those children scheduled for their second or third hydrotherapy procedure were eligible. The second or third hydrotherapy session was chosen due to control for consistency of child life specialist presence during the hydrotherapy procedure, level of desensitization of burn wounds, and any pharmacological interventions that may have occurred in pre-hospitalization. Exclusion criteria included patients with cognitive or developmental delays, non-English speaking, and visual or physical impairments (i.e., burns located on the hands and forearms, burns around the eyes, broken arms, blindness) that would preclude the use of a tablet. Exclusion criterion also included children who were undergoing hydrotherapy with pre-medication administered orally. Demographic information (age, gender, and ethnicity), injury information (percentage of total body surface area affected-TBSA), and treatment (hydrotherapy bath number) were collected for statistical analyses purposes. Sample size calculations determined 15 participants were needed in each group for between group analysis (with a power of 0.65, alpha of 0.05, and 0.66 point reduction in pain using the 5 point FACES scale).

2.2. *Procedure*

Informed consent from a legal guardian and assent from age appropriate patients (ages 7–12) was obtained by the child life specialist. Waiver of assent was obtained by legal guardians for children between 4 and 6 years old. Once enrolled in the study with consent and assent, participants were randomly assigned using random sample numbers generated by computer software to one of two groups. One half of the participants

($n=15$) were assigned to the tablet distraction group. The second half of the participants ($n=15$) were assigned to the control group. A researcher, not involved in the recruitment or data collection, completed the randomization and provided the child life specialist the assigned intervention for participants. The child life specialist and nurses were blind to the randomization process but not the randomization outcome.

Participants underwent a hydrotherapy treatment, and the assigned intervention (tablet distraction or control) took place during the hydrotherapy procedure. Hydrotherapy at Children's of Alabama Burn Center takes place in a treatment room on a daily basis for non- or pre-surgical patients. Once a patient arrives to the treatment room, the nurse administers intravenous medication for pain management, old bandages are removed, and the patient then moves to the hydrotherapy table for the debridement and cleaning of burns. Standard dosing for intravenous (IV) pain medication in the Burn Center is consistent among all inpatients. IV orders for Morphine and Versed are included in new admission order sets in the Burn Center specifically for premedication for hydrotherapy by the prescriber. It is a single dose of each medication for pain control during hydrotherapy. The standard dose for IV Morphine is 0.1 mg/kg/dose, and the standard dose used for IV Versed is 0.05 mg/kg/dose with a maximum dose of 2 mg. All medication doses are weight based and medication ranges or titration orders are not permitted.

The data collection began at the start of the debridement and cleaning process. At Children's of Alabama, the hydrotherapy table is a stainless steel table with solid, raised side rails. There is a removable cushion along the table's surface for comfort. The water source, or water line, is an adjustable spray hose that is installed in the ceiling of the hydrotherapy room. The water temperature is regulated from a control panel on the wall of the hydrotherapy room. When communicating with children about the hydrotherapy procedure, the medical team refers to the procedure with age-appropriate terminology such as "special bath to clean burns".

For the debridement and cleaning process, a nurse uses water soaked gauze to wipe over the burn areas to remove dead skin cells. Open areas are cleaned first, and then, the nurse cleans healed and non-burned areas using wet gauze. Strategies, such as allowing children choices (i.e., side of the body to be cleaned first), wetting the gauze rather than spraying water, and avoiding jumping around to multiple areas at one time, are implemented to increase comfort in the patient. In summary, physical cleaning of the wounds is done in a systematic, logical manner as to minimize pain and trauma.

The medical team at Children's of Alabama utilizes the one voice approach. This means they practice only one medical team member speaking at one time. Therefore, each medical team member has a role. Psychosocial support is provided by the child life specialist, and the nurses complete the hydrotherapy procedure. Nurses' interactions with the patients include providing information about the procedure (i.e., it is time to lift your arm or it is time to move to the table) or continuing rapport. As part of the one voice approach, the nurses allow the child life specialist the role of distraction and avoid overstimulation or competing with the child life specialist. This approach minimizes the potential for interference regarding examining the effectiveness of the tablet distraction provided by a child life specialist.

After the hydrotherapy procedure, the nurse reported observed behaviors of pain. Pain was self-reported by the participant. The child life specialist recorded the length of the procedure as well as observed behaviors of emotions indicative of anxiety. To minimize potential bias, the nurses and child life specialist had individual scoring sheets.

2.3. *Interventions*

The tablet distraction group received child life supported distraction using a computer tablet (iPad by Apple Inc.) during their hydrotherapy procedure. The child life specialist assessed the participant's development, procedure, and needs and then engaged the child in tablet distraction by providing options for activities (e.g. applications) meeting the assessment. Participants were allowed to choose an application of their choice from options presented by the child life specialist. Tablet applications used included developmentally appropriate interactive games (i.e., car racing, bubble popping, cupcake making, etc.). The tablet applications always had an interactive component, increasing involvement in the tablet distraction. Participants were provided tablet distraction support throughout the duration of the hydrotherapy procedure. Distraction support included the child life specialist initiating and continuing attention on the tablet, playing for the participant in periods when children displayed difficulty leading the play, providing verbal support, praise, and comfort, as well as promoting breathing and other pain management techniques. In other words, the child life specialist utilized psychosocial support while implementing the tablet distraction.

The control group received standard care, including support by a child life specialist with no added distraction tools. Support by a child life specialist ranged based on the patients' needs with potential support including developmentally appropriate explanations, emotional support, and pain management techniques (i.e., breathing exercises). The child life specialist offered psychosocial support to the children in the control group but did not implement any form of distraction.

2.4. *Outcome measurement*

2.4.1. *FACES pain rating scale [19]*

The FACES pain rating scale is a self-reported measure assessing children's level of pain during a procedure [19]. It is a valid and reliable scale that is appropriate for assessing pain in children 3 years and older [19]. After the hydrotherapy procedure, children were asked to self-report their pain using the scale of six faces, ranging from very happy, no hurt (0) to hurts as much as you can imagine (5). Descriptors for each face were provided. For example, descriptions included: face zero is very happy because there is no hurt, face three hurts even more than a little, and face five hurts as much as you can imagine. After describing each face, the nursed asked the child, "Which face showed how much hurt you had during the bath?"

2.4.2. Nurse's reports

To assess the nurses' perception on the use of computer tablet distraction during hydrotherapy, the nurses reported their observation of the participant's pain. Immediately after the procedure, the nurse was asked to rate the overall level of pain experienced based on their observation using a scale of 0 (no pain) to 5 (very high pain).

2.4.3. Children's emotional manifestation scale (CEMS) [20]

The CEMS is a valid and reliable observation scale used to assess children's emotional response to medical procedures [20]. In the current study, participants' emotional responses were assessed using the CEMS before the hydrotherapy, during the hydrotherapy, and after the hydrotherapy, as a measure of their anxiety at each of these points. Five categories of behaviors, including facial expression, vocalization, activity, interaction, and level of cooperation, were observed. These five behaviors were assessed because they were valid indicators of behaviors that are suggestive of children's emotions [20]. Each behavior lists five descriptions of emotional behaviors that vary on intensity from 1 to 5, with the higher suggesting the appearance of more negative emotions. For example, the behavior of activity has the following levels: calm (1), annoyance (2), irritable (3), restlessness (4), and agitation (5). The child life specialist selected the description that best matched the child's behaviors before the hydrotherapy, during the hydrotherapy, and after the hydrotherapy. Scores were totaled, with possible scores at each time point ranging from 5 to 25, and higher score suggest the child displayed more negative emotions (i.e., anxiety) [20].

2.4.4. Length of time

The child life specialist noted the length of time for the hydrotherapy procedure. The procedure started when the participants moved to the hydrotherapy table and the nurse began cleaning the burn wound, and it ended when the nurse finished the bath of the non-burned areas and prepared to return the participants to the stretcher. Identifying a way to determine a start and end point allowed for consistency in measuring length of time across participants. Assessing length of time allowed the research team to determine if tablet distraction provided by a child life specialist increased the length of the procedure.

2.5. Statistical analysis

All statistical analyses were conducted using SPSS 24.00™. To begin, independent samples *t*-test and Chi-Square tests were completed to determine if there were any significant differences between groups on the descriptive and background characteristics. Both outcomes measures for pain (self-reported and observed) were ordinal and not normally distributed; therefore, Mann–Whitney U tests were run to determine if there were differences in self-reported pain and observed pain between the tablet distraction and control groups. A Spearman's rank-order correlation was run to determine the relationship between the self-reported pain and the nurses' observed pain. Differences between groups on presence of emotional distress during the procedure (CEMS) and length of time were completed using independent samples *t*-test. A repeated measures ANOVA with time as the repeated measure (i.e., before the hydrotherapy procedure, during the hydrotherapy procedure, and after the hydrotherapy procedure) and group assignment (i.e., tablet distraction provided by a child life specialist, control) as the between groups variable was conducted on the children's emotional response to the procedure (CEMS). Mauchly's test indicated the assumption of sphericity for children's emotional response had been violated, $\chi^2(2)=7.46$, $p<0.05$; therefore, the degrees of freedom were corrected using the Huynh–Feldt estimates of sphericity (E=0.88) for that variable. Paired samples *t*-tests were conducted for post hoc comparisons within groups.

3. Results

3.1. Sample

The study included 30 pediatric patients undergoing hydrotherapy as a standard treatment for burns. The consort diagram (Fig. 1) illustrates 31 children were assessed for eligibility. One eligible participant was excluded because the participant's parent declined participation. Thirty participants were randomized, received the allocated intervention, and were included in analysis. Fifteen participants were included in each group in the analyses.

Descriptive and background characteristics for children, including means, standard deviations, and group distributions are presented in Table 1. The participants ranged in age from 4 to 12 years ($m=7.47$, SD=2.54), with a majority being male (63%; $n=19$) and white (47%; n=14). The participants' mean TBSA was 7.85 (SD=7.8), and 60% ($n=18$) of the participants were undergoing their second hydrotherapy procedure with the remaining undergoing their third one. At baseline, no significant differences were found between groups for age, gender, ethnicity TBSA, and hydrotherapy bath number, suggesting both groups were similar in these descriptive characteristics (see Table 1).

3.2. Pain

Children's pain was assessed through self-reports and nurse observations. Mean differences in pain scores for each group are shown in Table 2. Self-reported pain for the tablet distraction group and the control group was not statistically different, $p=0.24$. Observed pain for the tablet distraction group ($Mdn=12.47$) was significantly less than the control group ($Mdn=18.53$, $U=67.00$, $p=0.48$). In addition, there was a moderate, positive correlation between self-reported pain and observed pain, ($r_s(18)=0.36$, $p=0.052$).

3.3. Emotional response

The pediatric burn patients' emotional response was assessed through the child life specialist's observations before the procedure, during the procedure, and after the procedure, using the CEMS. Mean differences in emotional response scores for each group are shown in Table 2. Emotions were significantly higher for the control group compared to the tablet distraction group during the hydrotherapy procedure

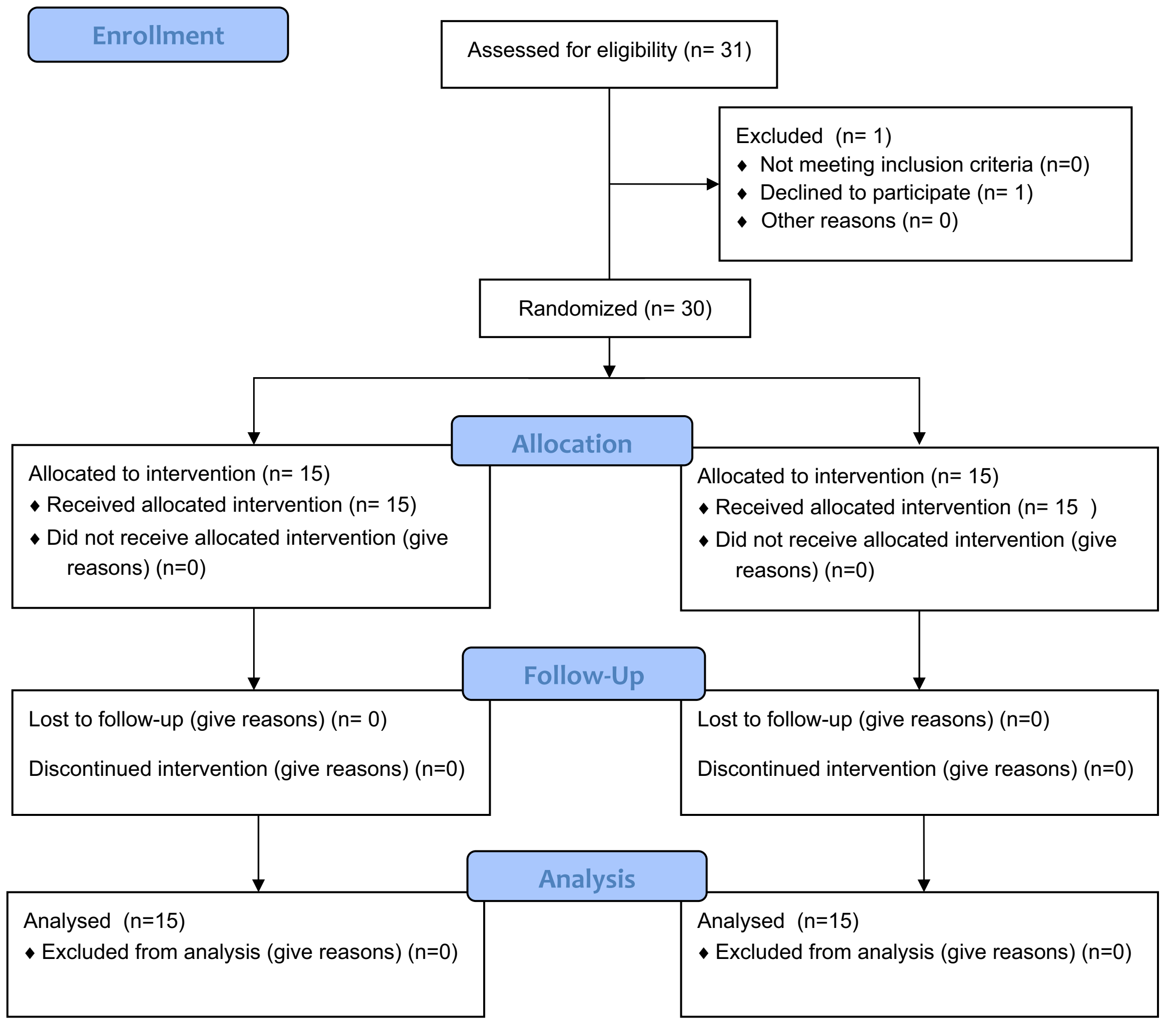

Fig. 1 – CONSORT flow chart.

($p = 0.001$) and after the hydrotherapy procedure ($p < 0.005$). Higher scores suggest more anxiety and negative emotions, indicating that children in the control group displayed more anxiety during the procedure and post-procedure than did the tablet distraction group.

A repeated measures ANOVA found a significant interaction between time and type of activity on children's emotional response to the procedure, $F(1.76,49.16) = 11.64$, $p < 0.001$. In the control group, children's emotional response significantly increased from baseline to during the

Table 1 – Baseline demographic characteristics.[a]

	Control ($n = 15$)	Tablet distraction ($n = 15$)	Total ($n = 30$)	t	χ^2	p-Value
Age, years	7.13 (2.77)	7.80 (2.34)	7.47 (2.54)	1.67		0.11
Gender					1.29	0.26
Male	11 (73.3%)	8 (53.3%)	19 (63.3%)			
Female	4 (26.7%)	7 (46.7%)	11 (36.7%)			
Race					1.31	0.52
White	8 (53.3%)	6 (40%)	14 (46.7%)			
Black	5 (33.3%)	8 (53.3%)	13 (43.3%)			
Other	2 (13.3%)	1 (6.7%)	3 (10%)			
Hydrotherapy number					0.56	0.46
2	10 (66.7%)	8 (53.3%)	18 (60%)			
3	5 (33.3%)	7 (46.7%)	12 (40%)			
TBSA (%)	6.43 (7.52)	9.17 (10.25)	7.80 (8.95)	0.83		0.41

[a] Data are means (SD) and numbers (%).

Table 2 – Pain, distress and outcome measures.

	Control (n=15)	Tablet distraction (n=15)	p-Value
FACES	3.20 (1.78)	2.53 (1.64)	0.29
Nurse reports of pain	3.73 (0.88)	2.93 (1.03)	0.03[b]
CEMS pre-procedure	9.47 (3.34)	7.67 (4.35)	0.21
CEMS during procedure	17.93 (4.83)	10.93 (4.96)	0.001[b]
CEMS post-procedure	12.00 (4.00)	7.33 (3.52)	0.002[b]

[a]Data are means (SD). FACES and nurse reports of pain utilized 5 point scales, with higher scores indicating more pain [19]. CEMS scores ranged from 5 to 25, with higher scores indicated increased anxiety [20].
[b] Statistically significant.

hydrotherapy ($p<0.001$), children's emotional response significantly decreased from during the hydrotherapy to after the hydrotherapy ($p<0.001$), and a significant difference was found between baseline emotions and post-procedure emotions with post-procedure distress being higher ($p<0.05$) (see Fig. 2). Higher scores reflect behaviors indicative of negative emotions and anxiety. Therefore, children in the control group appeared to display more anxiety during the procedure, and these behaviors calmed somewhat post-procedure. However, although emotions indicating anxiety lowered after the procedure, children in the control group did not return to baseline in regards to their emotional response to the procedure. In the tablet distraction group, children's emotions significantly increased from baseline to during the hydrotherapy ($p<0.001$), children's emotions significantly decreased from during the hydrotherapy to after the hydrotherapy ($p<0.001$), and a significant difference was not found between baseline emotional response to the procedure and post-procedure emotional response to the procedure (p=0.57). These findings suggest that children in the tablet distraction provided by a child life specialist group did return to their baseline level of anxiety.

3.4. *Length of procedure*

For the tablet distraction group, the length of the hydrotherapy sessions ranged from 5 to 17 min, with a mean time of 10.33 (sd=3.37). The length of the hydrotherapy session for the control group ranged from 5 to 20 min, with a mean time of 10.93 (sd=4.13). An independent samples t-test found no significant difference between groups for the length of the hydrotherapy procedure (p=0.66).

4. Discussion

The current study examined the effectiveness of tablet distraction provided by a child life specialist for pediatric burn patients undergoing hydrotherapy. Burn treatment procedures, such as hydrotherapy, are painful for pediatric patients, and the nonpharmacological method of distraction is thought to help focus patients' attention on a more neutral stimulus and away from the procedure or pain. It was expected that tablet distraction provided by a child life specialist would be an effective distraction tool as seen through a decrease in self-reported pain and observed pain and anxiety during the procedure. The results partially supported this hypothesis with the children receiving tablet distraction from a child life specialist demonstrating less observed pain and anxiety during hydrotherapy than the control group. However, a significant difference in self-reported pain was not found.

Nurses reported less observed pain in the children receiving tablet distraction provided by a child life specialist during the hydrotherapy procedure. Although not significantly, children in the tablet distraction provided by a child life specialist also self-reported less pain. The finding of decreased observed pain is a positive one considering pain is often thought to be undertreated in children receiving wound care and that analgesics alone may not be sufficient to treat the pain of these patients [3,4]. Such findings suggest that tablet distraction provided by a child life specialist may be an effective nonpharmacological method used in adjunct with analgesics

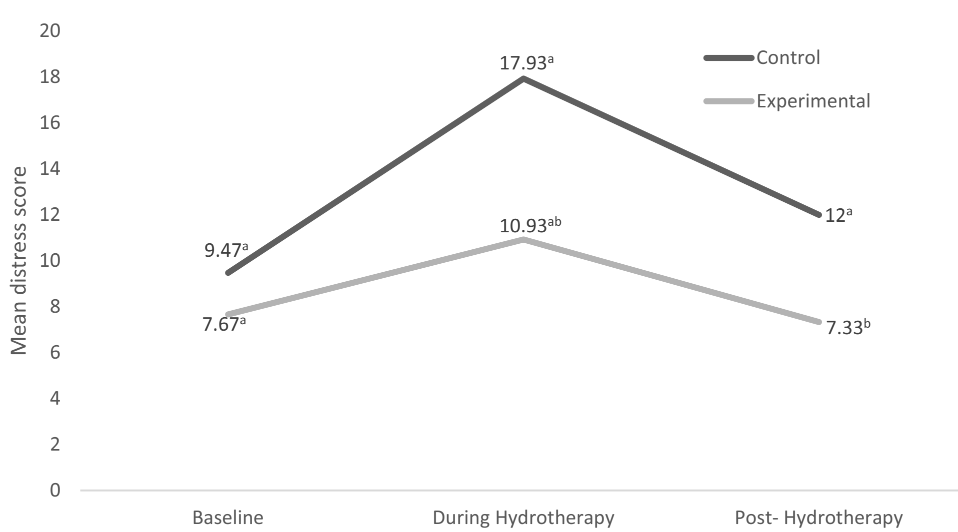

Fig. 2 – Researcher observation of children's distress (CEMS) within groups.
***Note.* Mean scores that have the same superscript are significantly different at the 0.05 level.**

for addressing the pain of pediatric burn patients during hydrotherapy.

These findings are not surprising considering a review of the literature on pediatric burn procedures shows similar results for other technology tools being used for distraction [6,10,12,13]. Other tools of technology, such as virtual reality and interactive gaming, have been shown to decrease pain during pediatric burn procedures [1,10,11]. Tablets are often used in this population during treatment procedures, but previous studies have not examined the effectiveness of tablets during such procedures. Thus the current findings contribute to the literature by providing preliminary evidence for an additional nonpharmacological tool to use in adjunct with analgesics with children undergoing burn wound care.

Anxiety during the procedure was assessed using behavioral observation of emotions because negative emotions and behavior responses are related to anxiety [20]. It was found that children who received tablet distraction by a child life specialist during hydrotherapy exhibited fewer negative emotions and behaviors than the children who did not receive tablet distraction. These findings suggest that tablet distraction provided by a child life specialist may decrease anxiety in pediatric burn patients during hydrotherapy. Also, after the procedure, children in the tablet distraction group were able to return to baseline emotions (i.e., emotional response prior to the hydrotherapy). Children in the control group did not return to their baseline emotions. They reported significantly higher scores on negative emotions and behaviors after the procedure compared to before the procedure, suggesting that these children were still anxious post-procedure. The finding that children in the control group exhibited more negative emotions and behaviors post-procedure than the tablet distraction group offers further support that the children in the control group continued to be anxious after the procedure. Overall, the findings found that tablet distraction provided by a child life specialist decreased anxiety during the procedure and helped children return to baseline post-procedure.

Decreasing anxiety is important because when anxiety is minimized children go into procedures with increased comfort, success, and control [21]. Decreasing pain and anxiety is important in pediatric burn patients because pain and anxiety increase the stress of a patient, and when a patient is stressed, healing can be impaired [2]. For example, stress is related to a slower healing process in patients undergoing different types of wound care [22–24].

Length of procedure was assessed, and a difference in the length of the hydrotherapy procedure was not found between groups. This means that utilizing tablet distraction provided by a child life specialist as a psychosocial intervention to minimize pain and distress did not lengthen the procedure. When selecting nonpharmacological approaches, procedures that are effective but do not increase procedure time are preferred as this helps the medical team maintain schedules and needs of the unit. The current findings offer support to the ability for tablet distraction provided by a child life specialist to minimize pain and anxiety but not increase the length of the procedure.

This study assessed tablet distraction provided by a child life specialist. A child life specialist is a member of the medical team who promotes the psychosocial coping of pediatric patients and their families by providing developmentally appropriate interventions and information about healthcare experiences. These members of the medical team have a minimum of a bachelor's degree with a focus on the development of children and families (i.e., child life, human development and family studies, early childhood development), experience observing and implementing the theories, approaches, and interventions of child life through a 480-hour hospital-based internship, and certification through the international certifying body of the Association of Child Life Professionals. Child life specialists assess the development and psychosocial needs of patients to design interventions appropriate to the individual patient.

In the current study, the child life specialists helped select a developmentally appropriate application for each of the participants in the tablet group, based off of the participant's development and interest. In addition, the child life specialist used her knowledge of psychosocial care to engage the children in the tablet and redirect them if needed. For example, during a period of high pain, a participant might look away from the tablet to breath and get through the pain. The child life specialist would have provided psychosocial support during such a time by encouraging breathing and then directing the child back into the distraction activity when the child's cues reflected he/she was ready to do so. Such interactions likely influenced the results. Therefore, the findings reflect the effectiveness of tablet distractions provided by child life specialists. Future studies are needed to further examine the effectiveness of tablet distraction alone versus tablet distraction provided by a child life specialist.

4.1. *Limitations*

It is important to note the limitations of this study. For one, the study included a smaller convenient sample; a larger sample size would have been more representative of the population. Second, due to the nature of distraction, one limitation was the inability to ensure that staff and participants were completely blind to the purpose of the study, increasing the potential for measurement bias. Bias was reduced by not informing the staff of the hypothesis of the study and minimizing the idea that tablets were the preferred choice. Next, pain was assessed by the participants and nurses after the procedure rather than during it. Reporting after the procedure could have impacted the accuracy of the pain assessments.

There was also potential bias in the use of raters who were involved in the study for the measures of observed pain and observed anxiety. The child life specialist who provided the tablet distraction also assessed the children's emotions using the CEMS. The nurse who provided the hydrotherapy assessed the observed pain of the patients. There is potential bias in this approach. Replication of the current study is needed with the use of a third uninvolved rater for observed pain and anxiety scores.

A final limitation is the generalizability of the findings. The current study examined the effectiveness of tablet distraction during a specific procedure (i.e., hydrotherapy) with a specific population (i.e., pediatric burn patients, IV analgesia). Therefore, these findings may not generalize to other procedures or populations. In addition, it is unknown if these findings generalize to the use of tablet distraction alone, without the support of a child life specialist. Future studies should replicate this study in other populations and examine the effectiveness of tablet distraction alone.

4.2. *Implications*

It is well documented that burn procedures are associated with significant pain in pediatric patients. In addition, there is a need for more than the standard pharmacological treatment methods for this population. The current study provides some support for tablet distraction provided by a child life specialist to improve pediatric burn patients' experiences because tablet distraction by a child life specialist minimized observed pain and anxiety for these patients. When pain and anxiety are minimized, patients are able to cope better with the pediatric experience and focus on healing. Therefore, burn clinics should consider implementing such strategies for their pediatric patients.

The current study also has implications for tablet distraction. Tablets are becoming a popular choice as a distraction tool for child life specialists and other healthcare professionals, but there is little evidence regarding their effectiveness. The current study contributes to the handful of studies that show support of tablet distraction and suggests that tablet distraction is an effective tool during medical procedures. Healthcare facilities and parents may consider the use of tablet distraction during pediatric procedures, and future studies should continue to examine tablet distraction during medical procedures.

5. Conclusion

To date, studies examining the effectiveness of tablet distraction in children undergoing treatment procedures for burns are rare. Additional nonpharmacological methods, such as tablet distraction, are needed for this population as pain is a continuous issue during wound care. It was hypothesized that children receiving tablet distraction by a child life specialist would have less pain and anxiety compared to children who did not receive tablet distraction while undergoing hydrotherapy. It was found that children who received tablet distraction provided by a child life specialist displayed less observed pain and anxiety compared to a control group. These findings suggest that tablet distraction provided by a child life specialist may be an effective tool for improving the pain and anxiety of pediatric burn patients during hydrotherapy.

Contributions of authors

Study concept and design: S. Burns-Nader, L. Joe, K. Pinion. Acquisition of data: L. Joe. Analysis and interpretation of data: S. Burns-Nader. Drafting of the manuscript: S. Burns-Nader, L. Joe, K. Pinion. Revision of the final manuscript: S. Burns-Nader, L. Joe, K. Pinion. All authors approved the final draft.

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Conflict of interest

All authors warrant that they have no conflict of interest to declare with respect to this study and the findings stated herein. Further, no sponsorship has been received from a proprietor, a distributor, or the manufacturer of the equipment used in the study.

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